CSE302

Introduction

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Data vs Information

Computer Data

- Data is a collection of values. Those values can be characters, numbers, or any other data type.
- Computer data is a bunch of 1's and 0's, known

as binary data.

- Computer data is processed by the computer's CPU and is stored digitally in files and folders on the computer's hard disk.

Example of Data:

EWU, Email:, Mr. X, [x@ewubd.edu,](mailto:x@ewubd.edu)

Dept. of CSE, Student

Information

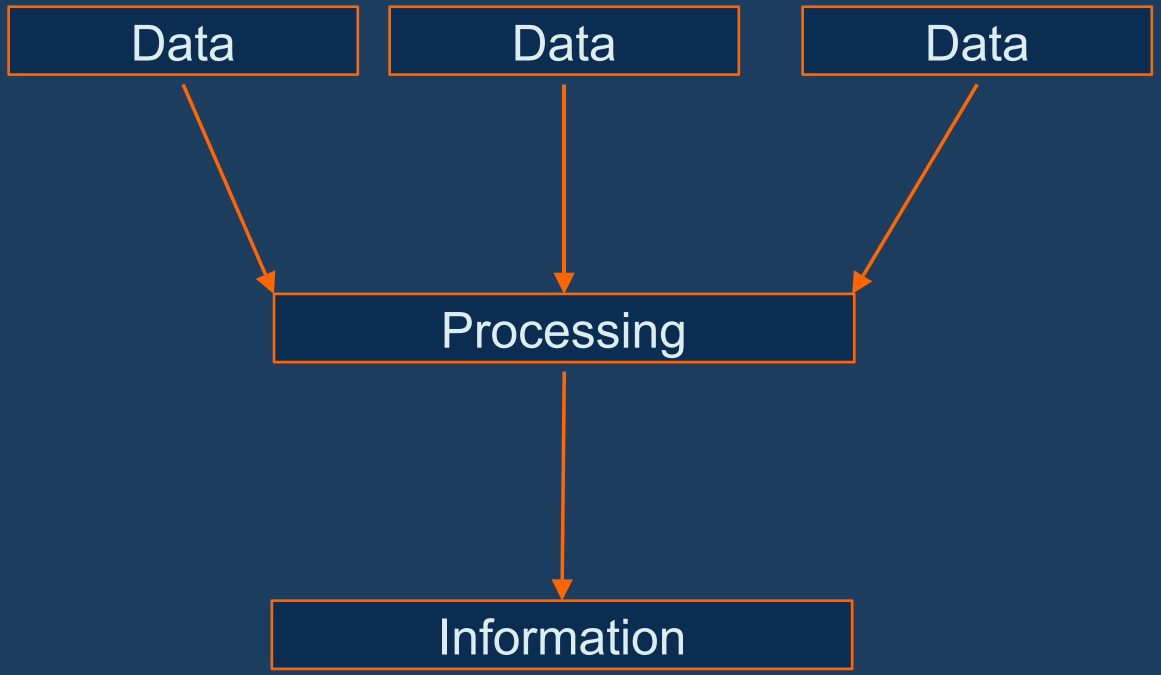
- When data are processed, interpreted, organized, structured or presented in a given context so as to make them meaningful or useful, they are called information.
- Data is raw material(i.e. bits of information) and Information is the product.

Example of Information :

Mr. X

Student
Dept. of
CSE
EWU
Email: x@ewubd.edu

Data vs Information



Data and Information

Raw vs Processed

Meaningful or not

Database?

- **Database:** The **collection of data** itself.
- **DBMS**: Software that **manages the database**.
- Before the concept of became common, data was managed in **traditional file systems**.
  - **Problem arises:**
    - Data Redundancy
    - Poor Data Security
    - No Query Support
- To overcome these limitations, **DBMS** was developed. It:
  - Security and access control via user permissions.
  - Powerful query: using SQL.
- **RDBMS**: It is a type of DBMS that stores data in a structured, tabular form — **using rows and columns** — and uses relationships between tables to organize data efficiently.

Why Database?

- Store
- Retrieve
- Update
- Delete

Why Database?

- Memory
  - Volatile (Register, Cache, RAM)
  - Non-Volatile (SSD, HDD)

Why Database?

EFFICIENT

INTEGRITY AND CONSISTENCY

SECURITY

REDUNDANCY

CONCURRENCY

Types of Databases

- Relational Database
- NoSQL Database

Where do we use Database?

Enterprise Information

Manufacturing

Banking and Finance

Educational Institutes

Where do we use Database?

Transportation

Telecommunication

Web Services

Document Database

View of Data

Data Models

Data Abstraction

Hides the complexity of data structures

View of Data

- A database system is a collection of interrelated data and a set of programs that allow users to access and modify these data.
- A major purpose of a database system is to provide users with an abstract view of the data.
    - Data models
        - A collection of conceptual tools for describing data, data relationships, data semantics, and consistency constraints.
    - Data abstraction
        - Hide the complexity of data structures to represent data in the database from users through several levels of data abstraction.

Data Models

- A collection of tools for describing
  - Data
  - Data relationships
  - Data semantics
  - Data constraints
- Relational model
- Entity-Relationship data model (mainly for database design)
- Object-based data models (Object-oriented and Object-relational)
- Semi-structured data model (XML)
- Other older models:
  - Network model
  - Hierarchical model

Relationship Model

Columns

Rows

ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

(a) The *instructor* table

Relationship Model

ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
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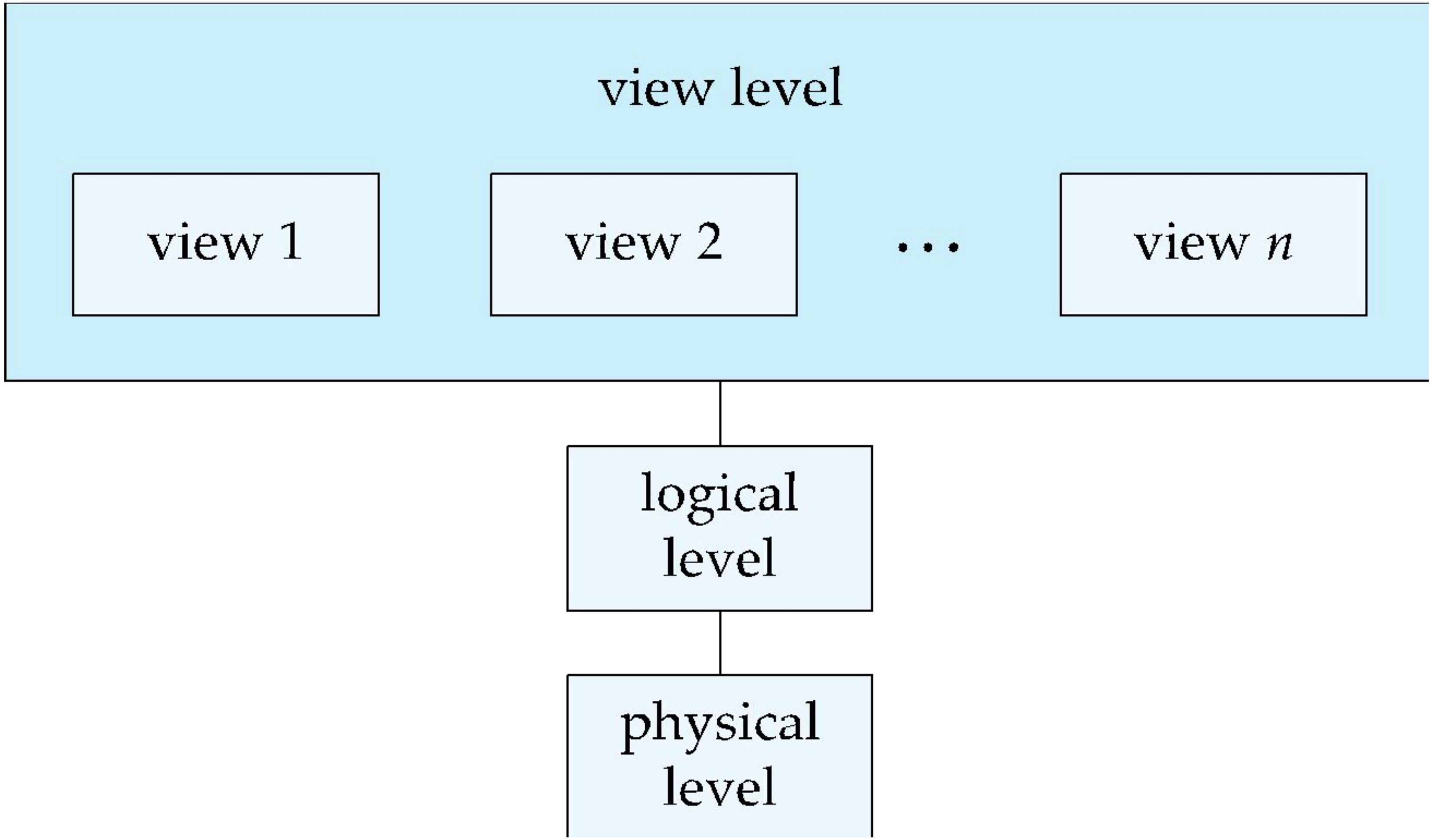
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

(a) The *instructor* table

dept_name	*building*	*budget*
Comp. Sci.	Taylor	100000
Biology	Watson	90000
Elec. Eng.	Taylor	85000
Music	Packard	80000
Finance	Painter	120000
History	Painter	50000
Physics	Watson	70000

(b) The *department* table

Abstraction



Levels of Abstraction

- **Physical level**: describes how a record (e.g., instructor) is stored.
- **Logical level**: describes what data stored in database, and the relationships among the data.

  type *instructor* = **record**

  ID : string;
  name : string;
  dept_name : string;
  salary : integer;

  end;
- **View level**: **Highest level of abstraction.** application programs hide details of data types. Views can also hide information (such as an employee's salary) for security purposes.

Data Definition Language (DDL)

- Deals with structure of the data table.

  create table *instructor* (
  ID **char**(5),
  name **varchar**(20),
  dept_name **varchar**(20),
  salary **numeric**(8,2))

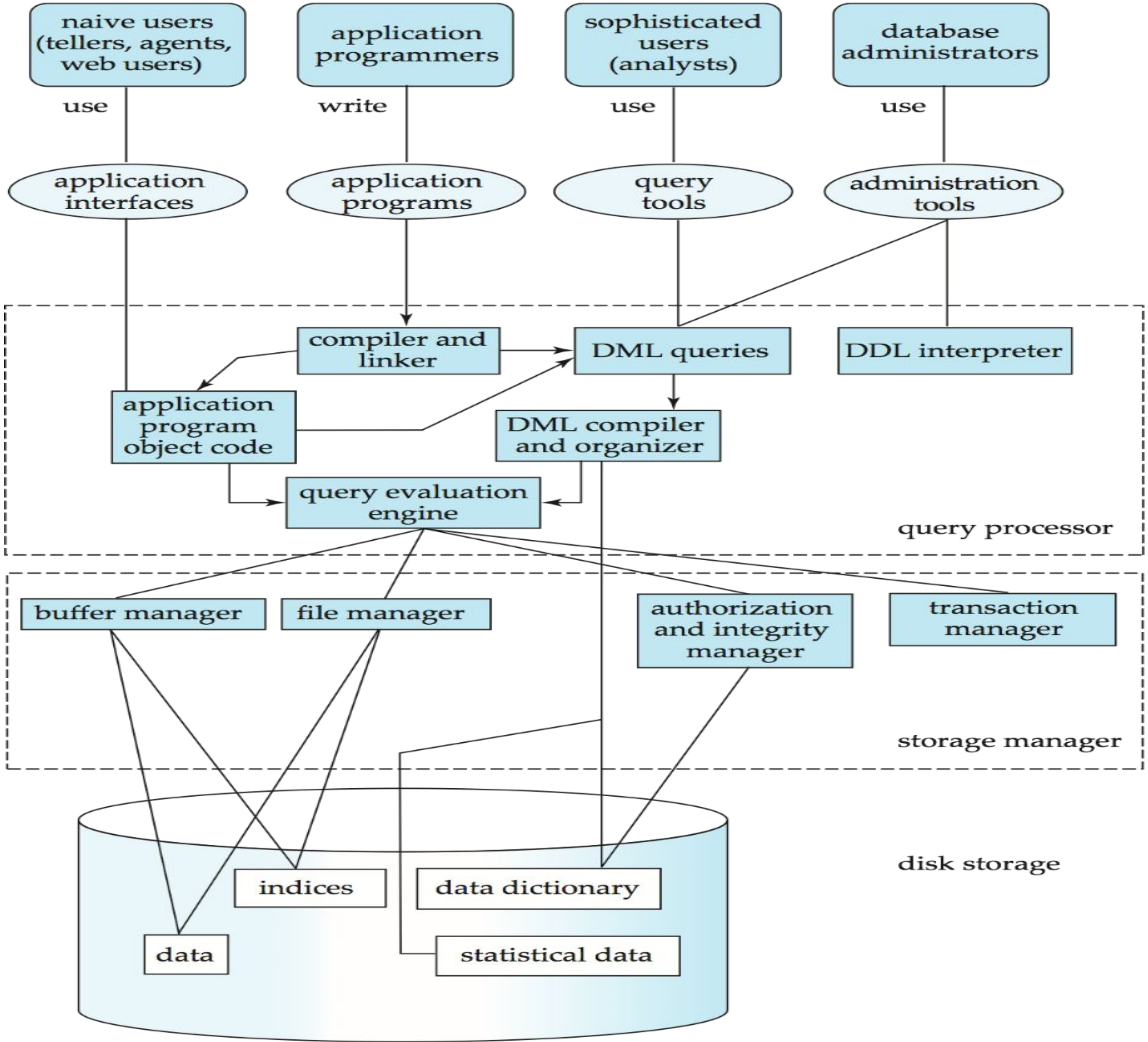
- DDL compiler generates a set of table templates stored in a ***data dictionary***
- Data dictionary contains metadata (i.e., data about data)
  - Database schema
  - Integrity constraints
    - Primary key (ID uniquely identifies instructors)
  - Authorization
    - Who can access what

Data Manipulation Language (DML) / Query Language

- For accessing and updating the data
- Two types
  - **Procedural DML** -- what data are needed and how to get those data.
  - **Declarative DML** -- what data are needed.
- Declarative DMLs are usually easier to learn and use.

Database System Structure

Database System Structure refers to the internal components of the DBMS, including the Query Processor, Storage Manager, and Disk Storage. It also defines the interaction of these components.



Database System Structure

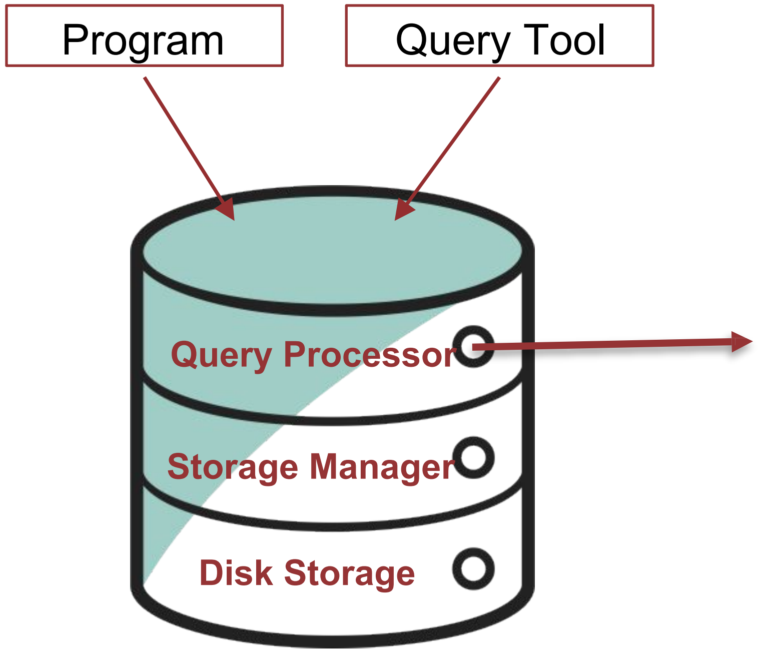
Database Users and Interfaces
Different types of users interact with the database through various tools and interfaces:

User Type	Interface Used	Purpose / Function
Naïve Users	Application Interfaces	Use predefined applications (e.g., ATM system, ticket booking). They do not write SQL queries.
Application Programmers	Application Programs	Develop programs using high-level languages with embedded SQL (e.g., Java, C++).
Sophisticated Users	Query Tools	Write complex queries using SQL or specialized tools for data analysis.
Database Administrator (DBA)	Administration Tools	Manage database structure, security, and performance using specialized administrative tools.

Database System Structure >> Query Processor

Query Processor

It interprets the requests (queries) received from the end user via an application program into instructions.

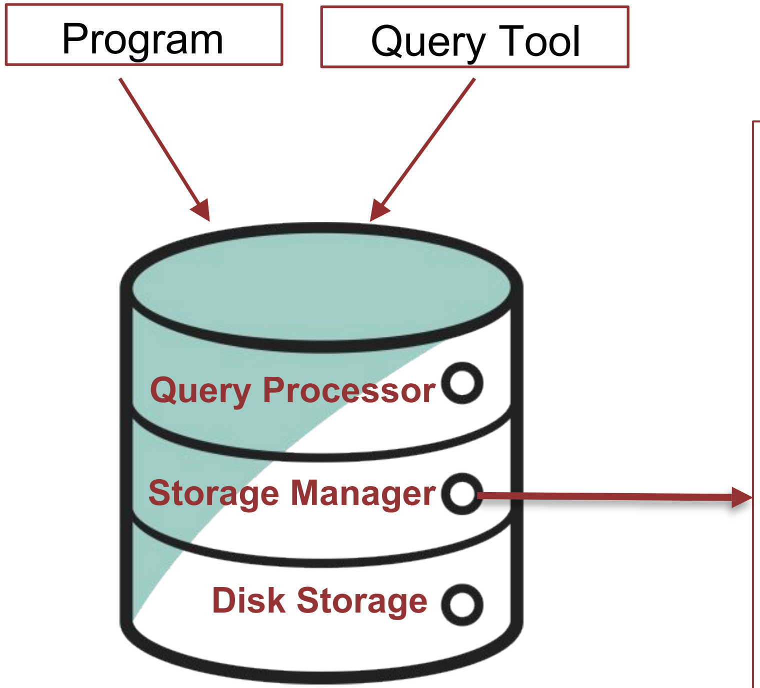


Query Processor

- **DDL interpreter**: Processes Data Definition Language (DDL) statements (CREATE, ALTER, DROP) and updates the **Data Dictionary** with new schema information.
- **Query evaluation engine**: executes low-level instructions generated by the DML compiler.
- **DML Compiler & Organizer:** Translates Data Manipulation Language (DML) statements (SELECT, INSERT, UPDATE, DELETE) into low-level instructions that the system can execute.
  - Also performs **query optimization** i.e. it picks lowest cost evaluation plan from a number of alternative evaluation plans.

Database System Structure >> Storage Manager

Storage Manager
It’s handles all interactions with the physical storage of data. It manages how data is stored on disk, retrieved efficiently, and maintained securely.

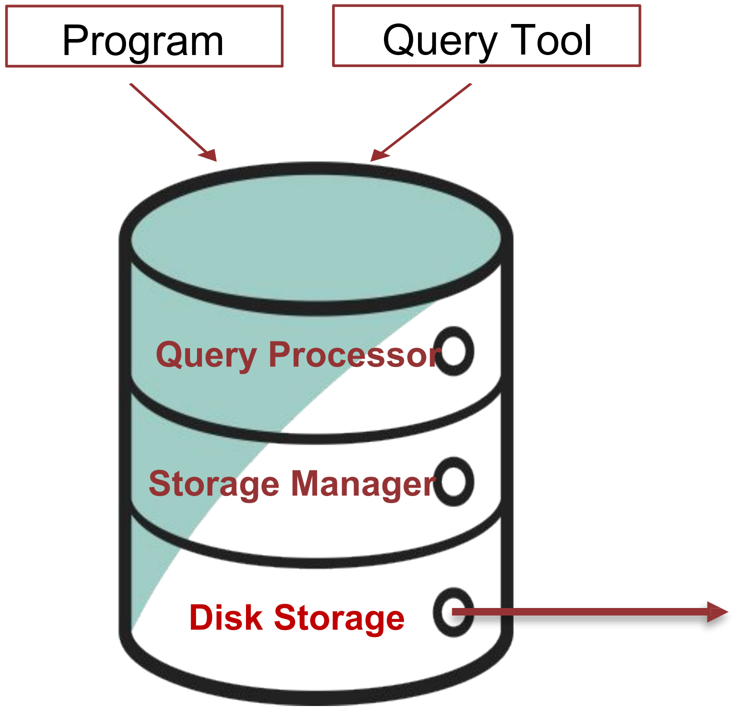


Storage Manager

- **Authorization and integrity manager** tests for the satisfaction of integrity constraints and checks the authority of users to access data.
- **Transaction manager** ensures that the database remains in a consistent (correct) state despite system failures, and that concurrent transaction executions proceed without conflicting.
- **File manager** manages the allocation of space on disk storage and the data structures used to represent information stored on disk.
- **Buffer manager** is responsible for fetching data from disk storage into main memory, and deciding what data to cache in main memory.

Database System Structure >> Disk Storage

At the bottom of the architecture lies the **physical storage**, where all actual data resides.



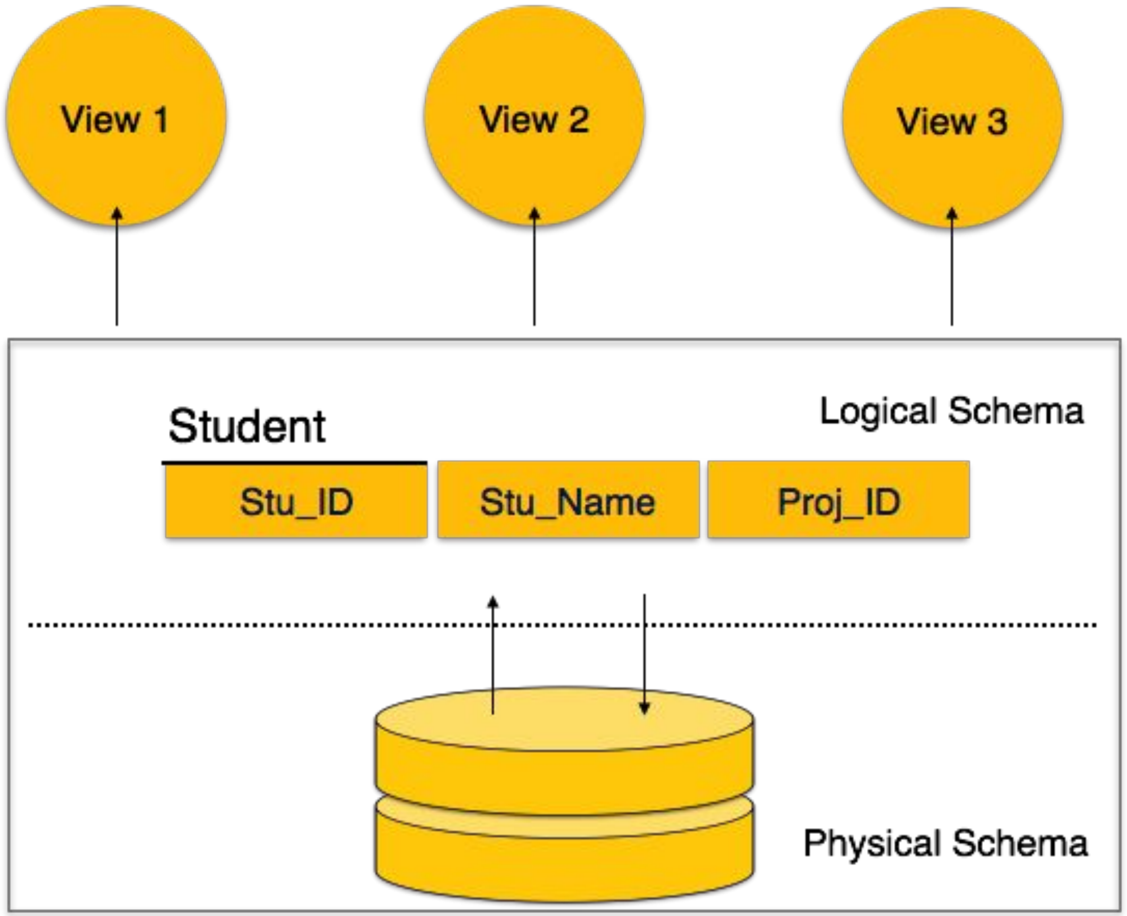
Disk Storage

- **Data :** The actual database content (records stored in tables).
- **Data dictionary**: which stores metadata about the structure of the database, in particular the schema of the database.
- **Statistical Data:** Maintains statistics (like table size, distribution) used by the Query Optimizer to choose efficient query plans.
- **Indices**, which provide fast access to data items. A database index provides pointers to those data items that hold a particular value.

Database Schema

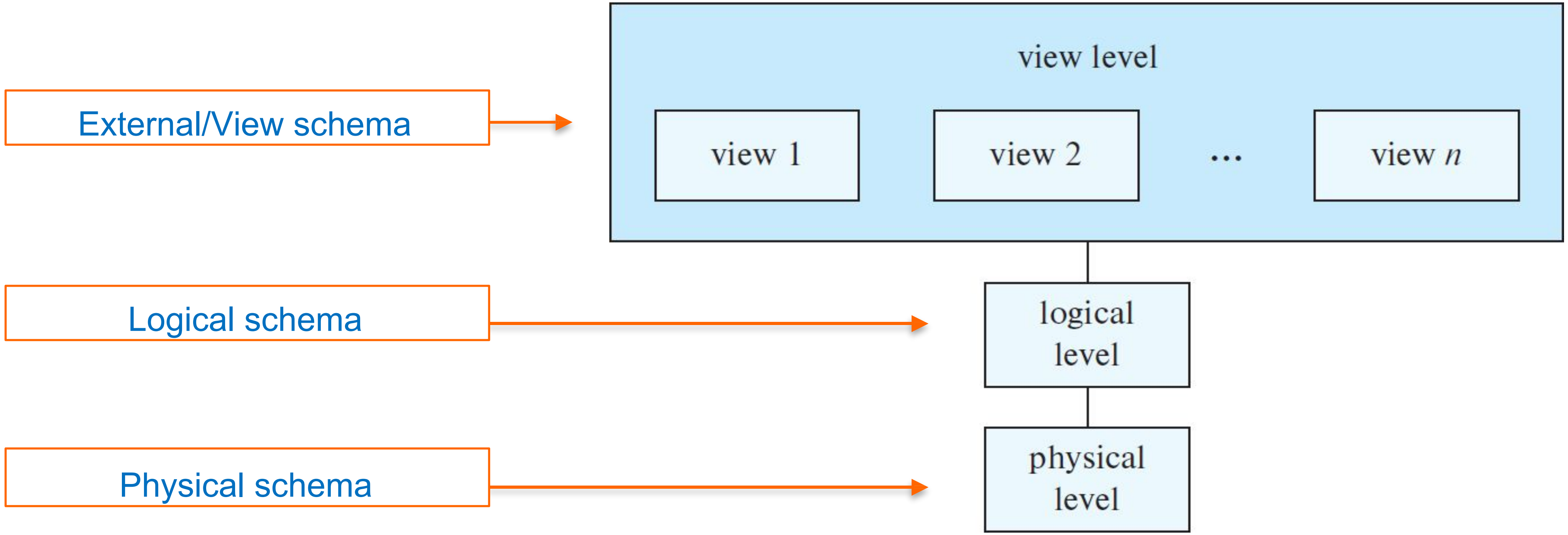
- **Database schema is the blueprint or structure that defines how data is organized in a database. It describes, what data tables exist, what columns each table has, relationship between tables, constaints like primary key or foreign key.**

- **Physical Database Schema** − It defines **how** the data will be stored in a secondary storage. This schema pertains to the actual storage of data and its form of storage like files, indices, etc.
- **Logical Database Schema** − This schema defines what data are stored and what is relation among them. Describe tables, fields and relationships.
- **View Database Schema –** Highest level of abstraction focusing on how user interact with database**.**



Database Schema

- Schema describes the overall design of the database at different levels.
- Each schema(logical) corresponds to a data model that is a collection of conceptual tools and languages for describing data, data relationships, data semantics, and consistency constraints.



SQL vs NoSQL Database

SQL	NoSQL
Relational Database	Non-relational Database
SQL databases use structured query language and have a predefined schema.	NoSQL databases have dynamic schemas for unstructured data.
SQL databases are vertically scalable.	NoSQL databases are horizontally scalable.
SQL databases are table based.	NoSQL databases are document, key-value, graph or wide-column stores.
• SQLite • PostgreSQL • Oracle • MySQL • Microsoft SQL server	• MongoDB • Cassandra

Relational Database >> MySQL

Student

student_id	name	age
1	ABC	25
2	DEF	20
3	GHI	22
4	JKL	27

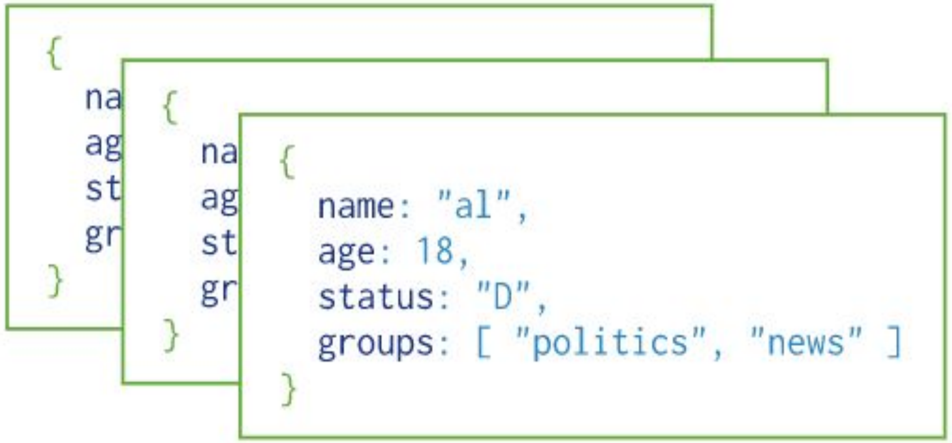
Course

course_id	name	faculty
1	Java	X
2	Python	Y
3	CPP	W
4	JS	M

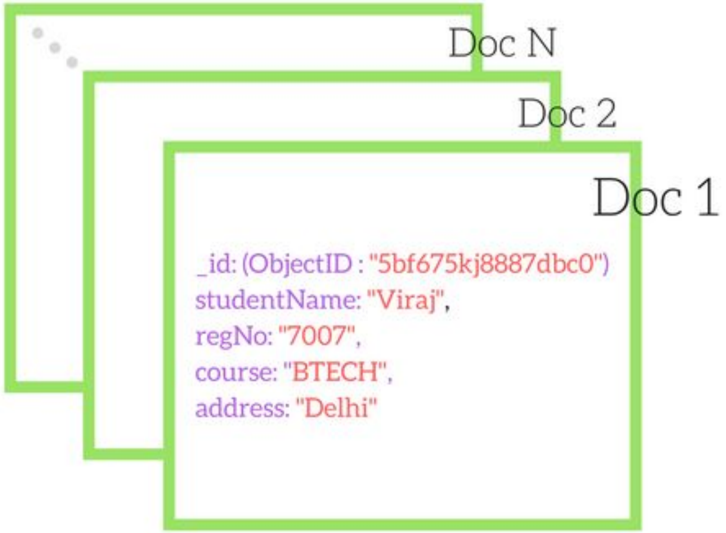
Student_Course

student_id	course_Id	marks
1	4	85
1	3	90
2	3	70
4	2	67

Non-relational Database >> MongoDB



Collection 1



Collection 2

DBMS VS RDBMS

Feature	**DBMS** (Database Management System)	**RDBMS** (Relational Database Management System)
Full Form	Database Management System	Relational Database Management System
Definition	A software for storing and managing data	A DBMS that stores data **in tables with relationships**
Data Storage	Data can be stored as files, key-value, or hierarchical	Data is stored in **tables (relations)**
Relationships	Does **not support** relationships between data	**Supports relationships** using **primary/foreign keys**
Normalization	Not supported or optional	**Enforces normalization** to reduce redundancy
Query Language	May not use SQL (or uses basic queries)	Uses **structured SQL**
Data Integrity	Low or manual enforcement	High integrity with constraints (e.g., NOT NULL, UNIQUE)
Transactions	Not guaranteed to support **ACID** properties	Supports **ACID** transactions
Multi-user Access	Limited	**Concurrent multi-user support** with locking and isolation
Examples	XML-based DB, File System, dBASE, Microsoft Access	MySQL, Oracle, PostgreSQL, SQL Server

Resources

- https://www.db-book.com/
- https://www.db-book.com/slides-dir/index.html